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Unlocking Stranded Reserves in Abu Dhabi Marginal Oil Fields Using Unprecedented Development Strategies

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Abstract

The development of marginal offshore fields is inherently challenging due to limited reserves and CAPEX-intensive projects. These challenges are compounded when resources are trapped across multiple stacked reservoirs and structures. This paper outlines the associated technical hurdles and innovative development strategies for stranded reserves in marginal structures, emphasizing the implementation of advanced reservoir monitoring technologies and adaptation of novel concept for facilities' sub-structure to set a precedent for early monetization of marginal offshore field developments in the region. This paper presents a breakthrough case study of Abu Dhabi Offshore Block-5 (OFFB-5), awarded to Pakistan International Oil Limited (PIOL) as block operator by ADNOC in 2021, where a novel, integrated, and cost-effective development concept was adopted to monetize three previously undeveloped discoveries. With first oil targeted by Q1 2028, this project exemplifies how technology, innovative reservoir modeling, and unconventional infrastructure solutions – such as a converted jack-up rig utilization as a Mobile Offshore Production Unit (MOPU) – can redefine the approach to marginal field development in the UAE. This paper shares the strategic, technical, and operational framework that enabled the transformation of seemingly non-viable discoveries into a feasible development opportunity, offering insights that can be replicated in other marginal offshore settings worldwide.

Introduction

In the United Arab Emirates (UAE), the focus has traditionally been on high-recovery, infrastructure-rich giant fields. However, the untapped potential of marginal offshore fields—characterized by limited reserves, structural compartmentalization, and capital-intensive development costs—demands new approaches. These marginal fields, namely Bu Dana, Al Bateen, and Al Manhal, were often overlooked due to their limited size and economic constraints, therefore require innovative approaches to become more viable. Production from marginal fields is a key initiative to achieve ADNOC's P5 Production Enhancement Plan.

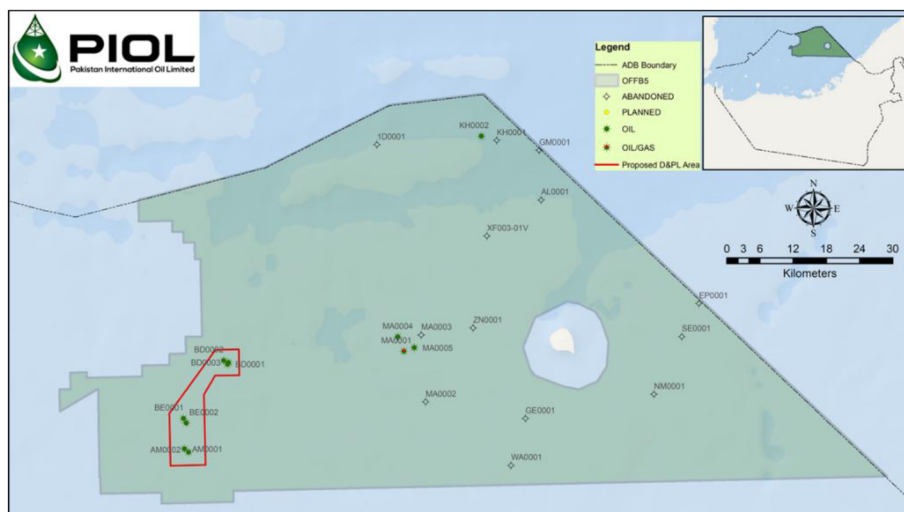


Figure 1—OFFB-5 Overall Map. The Red polygon shows existing discoveries which later formed the Production Concession boundary.

Field Overview and Challenges

OFFB-5 is an offshore exploration and development block spanning over 6,000 km². The block encompasses a geologically diverse landscape with complex structural and stratigraphic features with limited historical data. Within the block, three of five undeveloped oil discoveries were identified for development, each located within separate structures and characterized by multiple stacked reservoir intervals. Unlike conventional offshore fields in the region, the OFFB-5 discoveries are categorized as marginal due to their relatively limited reserves, remoteness to infrastructure, and dispersed resource distribution across multiple stacked reservoirs. These characteristics challenge the application of traditional development schemes. Additionally, the discoveries lie in an area devoid of existing production facilities or pipeline networks. Initial evaluations concluded that tie-back options to nearby producing assets were unfeasible due to capacity constraints and different development timelines. Deploying standalone fixed platforms would result in uneconomical capital expenditures.

From a subsurface perspective, the multi-reservoir setting introduces additional uncertainty in terms of connectivity and fluid flow dynamics. Operationally, the project faced time-bound development milestones, including a fast-track Field Development Plan (FDP) and a first-oil target by Q1 2028. These constraints required an unconventional, integrated approach encompassing advanced reservoir modeling, strategic facility concept selection, and fit-for-purpose technologies.

Methodology

To address the technical and economic challenges of marginal field development, a structured multidisciplinary workflow was adopted.

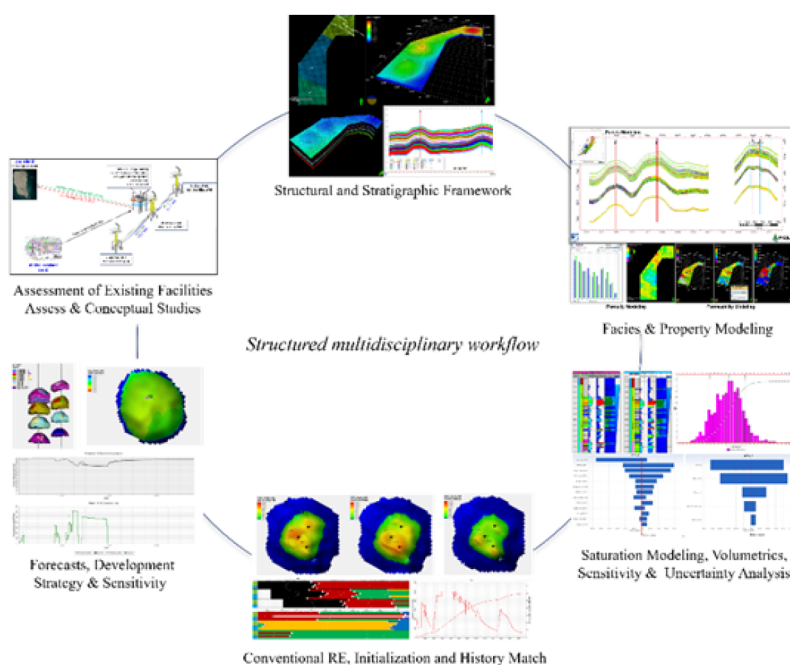


Figure 2—Structured Multidisciplinary Workflow.

Available Dataset used for Technical Evaluation

The datasets used for geological and geophysical evaluation for Bu Dana, Al Bateen and Al Manhal fields included log and core data for ~4 wells and Mega 3D Ocean Bottom Cable (OBC) 2015 PSTM cube with an area of ~414 sq.km.

Geophysical Study

The primary objective of the geophysical evaluation was to validate the structure and trap configuration for Bu Dana, Al-Bateen, and Al-Manhal fields. For this purpose, geophysical evaluation workflows were implemented based on the sequential integration of the available data (wells & seismic) through wells and seismic data QC, well to seismic ties, seismic interpretation/mapping, velocity modelling, depth conversion, and structural configuration uncertainty analysis.

Seismic interpretation shows small structures in terms of areal extent and vertical relief at the target reservoirs. A few faults were also interpreted at Bu Dana crust through the analysis of various seismic geometric attributes (e.g., variance, curvature, etc.). These faults have minimal throw mostly <10ms at the target reservoirs and are primarily oriented in the NW-SE direction, aligned with the regional trend. Confidence in seismic interpretation of Top Arab A to Upper Araej reservoir is generally good within the mapped structure area.

Petrophysical Study

Full field petrophysical interpretation was carried out for logs & core data conditioning followed by logs analysis for minerals, porosity and fluid saturation, sensitivity analysis on the output results (PHIE/ NTG and SW), core and log-based reservoir rock typing, permeability modelling and saturation height function that include prediction and computation of FWL.

All pre-requisite data were checked, that includes conventional logs, RCA analysis, and MICP analysis. In addition to the core and log data, additional sources referenced for the petrophysical study include, geological report, core lithology description, DST Test results to verify both productive reservoirs intervals, temperature information and to provide verification of interpreted water salinity, and the formation tops and well trajectory.

The detailed logs analysis included quality check of log and core data utilizing industry standard workflows. All anomalous core and log data were conditioned prior to processing of logs. Multi-mineral cross-plots and histograms supported mineralogical assessment and parameter selection. Petrophysical parameters were derived using calibrated inputs such as core-to-log depth shift, temperature profiles, mud properties, R_w & salinity, and mineral/porosity endpoints. Log results were then subjected to Monte Carlo sensitivity and uncertainty analysis to generate P10, P50, and P90 estimates. Final outputs were compiled in a summation report for every reservoir in each field.

Integrated Sedimentology Study

Detailed review and evaluation of available core and well data, regional studies/literature were carried out to understand the depositional setting, thickness variation, porosity trends, log characteristics and facies variation of all the proven reservoirs.

An integrated sedimentology, diagenesis and sequence stratigraphy study was conducted to understand the sedimentological and reservoir quality distribution within the sequence stratigraphic framework for Hith, Arab A, B, C & D, Diyab and Upper Araej. This study is conducted on four cored wells (along with physical core review sessions), utilizing complete log suites and a total of 250 thin sections description (including 65 new thin sections from 1B and 1C wells) and XRD, SEM and CL analysis on 16 samples.

The studied succession in these four wells is characterized by a total of 7 complete 3rd-order sequences, along with two 3rd-order regressive hemicycles. The Upper Araej Member to Diyab Formation (Hanifa Member), exhibits a series of large-scale shallowing-upward trends. These trends include the transition from mud-supported outer ramp deposits to mud-/grain-supported mid-ramp deposits, ultimately culminating in high-energy, grainstone-dominated inner ramp deposits. Moving upwards, the Jubaila Member (Diyab Fm) to Arab D Member shows a gradual shift from muddy outer ramp deposits to predominantly muddy intertidal deposits. The Arab C Member to Hith Formation is characterized by prevailing mud-/grain-supported inner ramp and mud-supported intertidal deposits, with a lesser contribution from supratidal settings. Large facies belts likely characterized the deposition of these formations, with slightly shallower settings observed in the northeast part of the study area due to the overall southwestward progradation of the carbonate system.

The integrated sedimentological, diagenetic, and sequence stratigraphic evaluation provided a robust framework to guide simulation input. Detailed core description, thin section petrography, and log calibration allowed for accurate facies definition, porosity–permeability trends, and reservoir quality distribution within a sequence stratigraphic context. By linking depositional setting and diagenetic overprint with petrophysical responses, the study enabled more realistic facies modeling, rock typing, and saturation functions. This integration reduced uncertainty in vertical and lateral heterogeneity, ensuring that static and dynamic models incorporated geologically consistent facies belts, porosity trends, and stratigraphic cycles, ultimately strengthening the predictive power of the reservoir simulation.

3D Reservoir Simulation Modeling

Built in accordance with ADNOC's modeling standards, the static model provided a robust geological and petrophysical foundation that enabled the identification and unlocking of stranded reserves in a marginal Abu Dhabi oil field through innovative development strategies.

Static Model. Static Model was developed as per the ADNOC standardized workflow. The workflow commenced with seismic interpretation and structural modeling, which included fault modeling to construct a geologically consistent fault framework, followed by pillar gridding to define the 3D grid structure honoring fault geometries. Model boundary selection is done considering the 400-600m distance from P10 contours at each interval of the deepest structure which is also the basis for segments division as well. Horizon modeling was carried out using interpreted seismic surfaces and well tops to define key stratigraphic boundaries. This was followed by vertical zonation based on depositional sequences and log signatures to preserve stratigraphic layering. Quality control (QC) checks ensured structural consistency,

and the model was upscaled from fine-scale layering to simulation-ready resolution while preserving critical heterogeneities.

Sedimentological interpretation and reservoir quality maps were provided through core and log based integrated sedimentological study, which were translated into 3D facies models using geostatistical methods constrained by well data and depositional trends. Petrophysical property modeling included porosity, permeability, and net-to-gross, guided by facies and rock types were carried out to populate the static model with realistic property distributions and to capture the reservoir heterogeneity.

The saturation height model was built which was based on a contact definition for each reservoir, and on a physical model of the saturation. Both were provided by an analysis of the petrophysical data, test, pressure gradient, logs available for each reservoir. The saturation model is computed using Leverett J-function compatible with ECLIPSE processes. The saturations were validated comparing the SW log from Archie laws with the SW grid property at the well and with cross plot analysis.

The base case model was used for volumetric estimation. Subsequently, the uncertainty assessment was carried out to capture the range of possible STOIIP outcomes by generating alternative static models representing P90, P50, and P10 cases. The full range 729 runs were done to assess the full range of volumes using a monte Carlo sampler driven by Latin-hypercube orthogonal sampling. The uncertainty range is intermediate ($\pm 30\%$), in line with the well data delineating the fields/size of the structures. Moreover, sensitivity analysis was performed to assess the most uncertain parameter. The analysis shows that the contacts were the most impacting uncertainty, followed by the permeability and porosity. Better definition of these parameters will strongly reduce the range of possible STOIIPs.

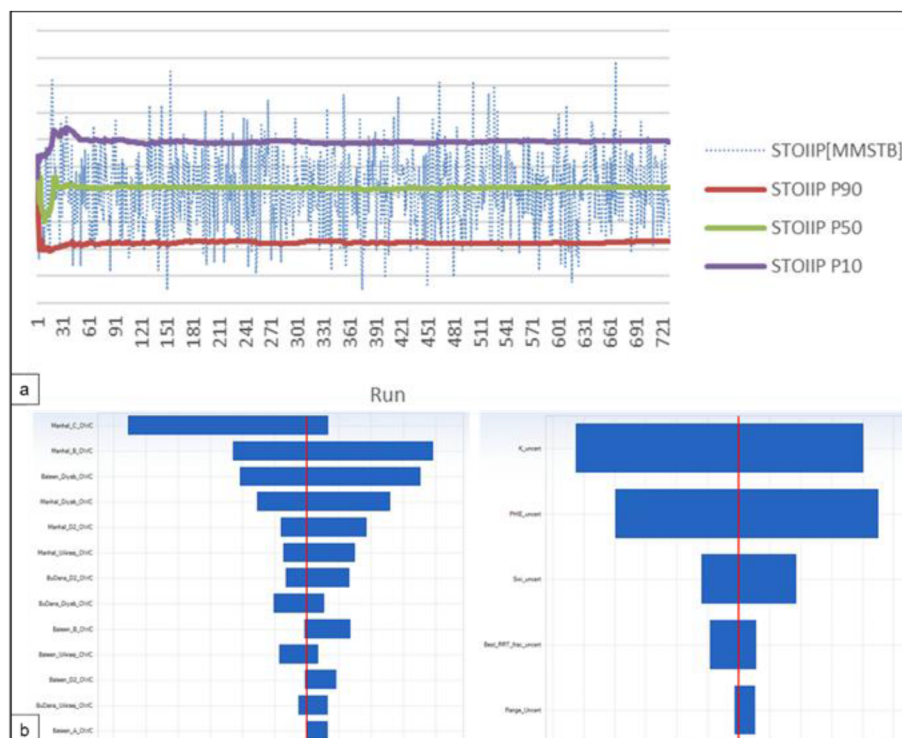


Figure 3—(a) Uncertainty analysis with 729 runs & (b) Tornado plots showing contacts & properties sensitivity in volumes.

Reservoir Engineering. Since the exploration wells on the discoveries were drilled in early 1980s with limited pressure data availability. Moreover, there was only one appraisal well on one discovery, while no appraisal data was available on other discoveries. There was no digital well test data available due to data being quite old. However, detailed PVT analysis for each reservoir was available which was utilized for PVT models and further analyses.

The available data (PVT, pressure, RFT, etc.) was digitized to be utilized for basic engineering analysis and interpretation, including but not limited to well test analysis (Productivity Index, Pressure Transient Analysis, skin, fault, mobility data and boundary analysis).

Dynamic Simulation. Full-field dynamic simulation models were developed for each stacked reservoir. These models incorporated petrophysical parameters, PVT data, and drive mechanisms. The models enabled optimization of well placement (including deviated and multilateral designs), informed pressure support strategies, guided tracer-based surveillance, and assisted in smart completion planning.

The initial grid quality checks (cell bulk volume, cell inside out and cell angles) were performed to initialize the model. Subsequently, using available production history during the DST/ production tests campaigns, the model was history matched by applying the "kh" and "skin" information obtained via pressure transient analysis. The history match model was utilized for various production strategies, including but not limited to sequential production, semi-commingled (bottom two reservoirs in one phase and upper three in 2nd phase) and commingled scenario, with sensitivities on artificial lifting mechanism employing Electrical Submersible Pump (ESP) and Gas Lift (GL). Moreover, sensitivities were also run on the number of production and injection wells and the timeline for the start of water flooding operations. The optimum scenario among the range of tested scenarios was identified in all the fields with water flood to begin right from the start and employing ESPs in all the producers to lift the oil and associated water. A total of 9 producers (3 in each field) along with 6 injectors (2 in each field) were the outcome of the optimum case, taking the total development well count to 15.

Appraisal Strategy. The 3 appraisal wells (one in each field) were planned in such a way that unlike the conventional offshore practice of abandoning the appraisal wells, the wells were suspended with Mud Line Suspension (MLS) system to re-enter the wells at later stage and use them as water injectors by side tracking from a shallow depth, hence saving several million dollars in injector well cost.

The entire development model has been an ingenious concept, planned for the first time in this region, to cater for the unorthodox challenges with an unorthodox approach.

Assessment of Existing Facilities

Considering the size of the field, independent facilities were not viable at an earlier stage. Therefore, a detailed Host Platform/Facility Assessment study was conducted to explore the possibility of utilizing the existing facilities and infrastructure to receive and process OFFB-5 production accounting for ongoing or future planned developments within those existing facilities. The scope included evaluation of existing facilities to address OFFB-5 oil and gas tie-ins, produced water disposal, and supply of water injection and power. The objective was to integrate OFFB-5 facilities with existing adjacent fields to maximize synergy. After comprehensive assessment, a standalone option was pursued primarily due to the lack of capacity in existing facilities and respective expansion plans, with timelines largely different from OFFB-5 first-oil target, among other constraints.

Assess and Conceptual Studies

Comprehensive Assess and Concept Selection studies were carried out to identify, assess, and explore multiple potential options to provide PIOL with a viable development concept to take forward into the next phase for the block development. The Assess study investigated the feasibility of adopting conventional standalone developments, existing third-party tie-ins, and fully stabilized oil processing via converted jack-up rig-based facilities (Mobile Offshore Production Unit – MOPU) targeting minimum footprint developments and fast-tracked solutions in view of the future developments in the area. Through robust technical definitions and assessment of the field development architecture, e.g., water injection strategy, artificial lift methods, power sourcing vs generation alternatives, etc., the Assess study resulted in compilation of a Concept Catalogue by utilizing the building-block approach. The Concept Catalogue was

taken forward to the next stage for further scrutiny, definition, and identification of feasible concept(s) as part of the Concept Selection study.

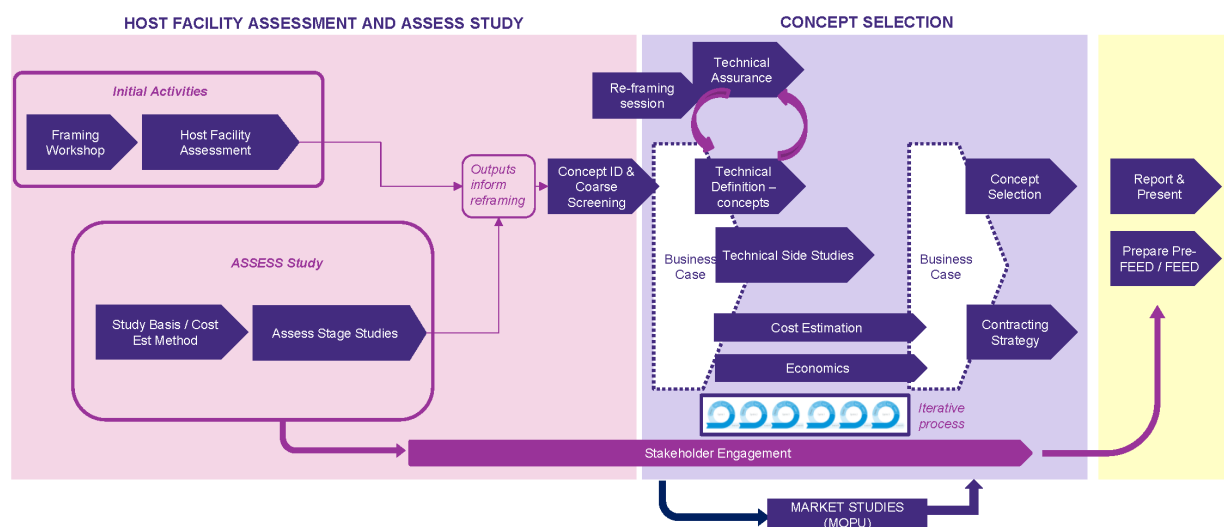


Figure 4—Overall Flow Chart of Surface Studies.

During the Concept Selection study, a data driven approach was applied by following the ‘MECE Principle’ through identification and assessment of a set of concepts which covered the full range of feasible development options ("Collectively Exhaustive") but were sufficiently restricted in number and differentiated that the required insights could be derived to support the decisions in the most time and cost-efficient manner ("Mutually Exclusive"). A direct comparison among each concept allowed key decisions to be informed without the need to develop every possible concept. As a result, the final recommended concept comprised of a hybrid of two or more concepts studied in detail by taking the most beneficial elements of each concept. A total of Nine (09) concepts were developed covering a wide range of variables. All options were technically defined along with the development of cost estimates, high-level schedules, risk and opportunity assessments, and UTC based economics. As a result, a MOPU based scheme was selected to ensure reduced CAPEX, fast deployment, and improved economics. The key elements are summarized in Table-1 below:

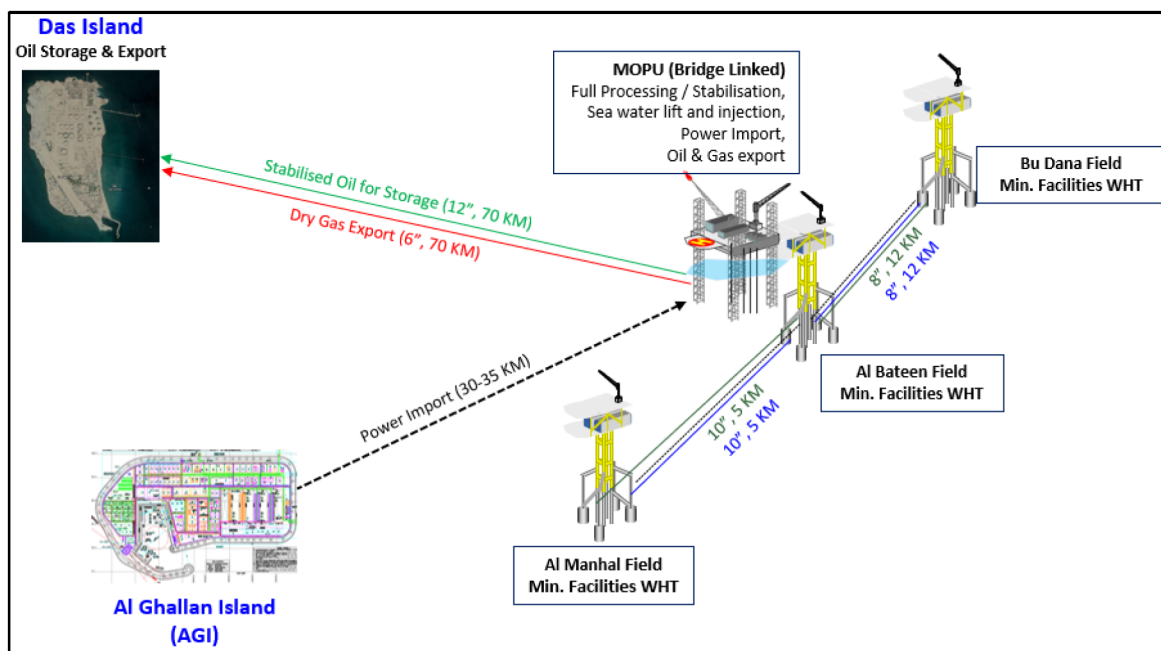
Table 1—Key Reservoir Properties

Formation	Arab A	Arab B	Arab C	Arab D2	Diyab	U. Araej
Porosity %	15-33	13-25	13-23	7-25	8-23	6-17
Horizontal Permeability, mD	10-20	64-78	34-60	19-130	96-174	16-37
Vertical Permeability, mD	4-9	9-11	7-12	9-53	41-72	3-6

Table 2—Key Elements - Selected Concept for the OFFB-5 Field Development Scheme

Key Elements	Details
Oil Fields	Bu Dana, Al Bateen, Al Manhal
Development Phase	Full Field Development (FFD) of the Three fields
Oil production Plateau Rate	25,000 barrels of oil per day
Wellhead Tower Type	Three Minimum facilities Conductor Supported Platforms (One per field)
Type of Fluid / Processing	Full Stabilization of Oil and Dehydrated Dry gas
Central Processing Facility	Jack-up type Mobile Offshore Production Unit (MOPU)
Host Facility for Oil Processing	Stabilized oil export to Das Island via new pipeline for storage and export
Host Facility for Gas Processing	Dehydrated Gas export to Das Island via new pipeline for further handling
Produced Water	Treatment and injection as 'Injection Water' after mixing with treated sea water
Water Injection	Sea Water lift pump and sea water treatment package
Artificial Lift method	Electrical Submersible Pumps (ESP)
Power Supply	Import from existing nearby Al Ghallan Island (AGI)

As per the selected concept, raw multiphase fluids from each wellhead tower will be transported to the Mobile Offshore Production Unit (MOPU) where all separation, processing, and treatment will be carried out. Fully stabilized oil and dry gas will be transferred to Das Island via new 70km long sub-sea pipelines. The required power will be supplied from nearby existing Al Ghallan Island (AGI) via a new 35km long cable. Each field will have its dedicated minimum facilities conductor supported wellhead tower with 9 slots per wellhead tower for production and water injection wells. The Injection water will be supplied from the MOPU via in-field pipelines where treated produced water will be mixed with treated sea water for onward injection through water injection wells located at each wellhead tower.

**Figure 5—OFFB-5 Overall Field Development Scheme.**

The selected option was then further defined through the Concept Design Study. The objective was to achieve refinement of cost ($\pm 30\%$), schedule (Level 2), as well as detailing technical aspects while focusing

on design optimization and robustness. Formal risk assessments and comprehensive HSE studies were performed, including Quantitative Risk Assessment (QRA).

Integrated Development Planning

A fast-track Field Development Plan (FDP) was submitted in Q4, 2023. The plan aligned subsurface strategy, phased drilling, and surface facility requirements with the Q1 2028 first oil target.

Results and Observations

Reservoir Simulation and Production Strategy

- EUR ranged around 35% of OOIP under optimized depletion in multiple stacked reservoirs and structures.
- The base case development plan included 9 producers and 6 injectors.

Infrastructure Constraints

- Adjacent facilities lacked processing capacity.
- Schedule misalignment and future long-term expansion plans prevented tie-back feasibility.

MOPU Concept Selection

- CAPEX reduced by around 35% compared to the conventional super-complex type phased installations.
- Enabled alignment with fast-track schedule.
- Improved UTC (21.5 \$/Bbl) and IRR (44%) ensuring commercial viability.
- First implementation of MOPU in UAE offshore sector.

Development Strategy

Subsurface

ESP is the optimum artificial lift system for the development of the reservoirs with application of water injection from day 1 to ensure reservoir pressure support and Voidage Replacement Ratio (VRR) maintenance and maximize oil recoveries from each reservoir.

For well intervention during production, Y-Tool was planned in the tubing to minimize production curtailment during well intervention, reservoir surveillance and data acquisition operations.

Development Philosophy

The selected MOPU concept reflected a shift to a modular, rapid-deployment infrastructure. Focused on capital efficiency, rapid execution, and subsurface adaptability, the Mobile Offshore Production Unit (MOPU) is a viable candidate for shallow water early production applications. MOPU can offer a cost-efficient solution for short life fields, compared with conventional jacket structure. MOPU has advantages in terms of mobility, ease of installation, hook up, commissioning, and decommissioning costs in comparison with a traditional jacket structure.

MOPU Concept Overview

Converted jack-up rig-based facilities will form the Mobile Offshore Production Unit (MOPU). In addition to the oil reception and processing facilities, the MOPU will have all associated systems, e.g., gas

sweetening, produced water treatment, sea water lift and treatment, pumping and compression, metering, utilities, accommodation, helideck, material handling, safety equipment, etc.

Prior to installation, the existing rig must be converted for the project specific site. It typically can take up to 16 to 24 months from project kick off to production depending upon the scope involved. This is a fast project schedule when compared to conventional super complex production platforms. MOPU installation typically requires a jack-up installation. The vessel can be towed, piggybacked, or self-propelled to the operating location. Upon arrival, the legs of the vessel are lowered from the buoyant hull until they reach the seabed. Then extension continues until the vessel is above the water line and reaches the chosen height.

Execution Timeline

Table 3—Key Milestones

FDP Submission	Q4-2023
FEED and FID	Q1-2026
Engineering, Procurement, Rig Conversion	Q4-2027
Installation and Commissioning	Q1-2028
First Oil	Q1-2028

Conclusions

This paper demonstrates how unconventional infrastructure and integrated subsurface planning can transform stranded reserves into economically viable project. Key lessons-learned include combined reservoir simulation modeling of Three (03) marginal fields, each having Five (05) different reservoirs, which resulted in considerable timesaving while ensuring overall synergy. This approach combined with the adaptation of novel lean sub-structures for CPF and Wellhead Towers, i.e., converted jack-up MOPU-based rig and conductor supported platforms respectively as alternatives to conventional sub-structures, offered new pathways for unlocking potential in OFFB-5. Close coordination and timely engagement with all stakeholders accelerated the review and approval process for field development. The integrated project management approach and successful implementation of this strategy established a benchmark for future marginal offshore field developments in the region. This approach is broadly applicable to other marginal fields facing similar challenges.

Acknowledgements

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Nomenclature

ADNOC	Abu Dhabi National Oil Company
CAPEX	Capital Expenditure
CPF	Central Processing Facility
ESP	Electrical Submersible Pump
EUR	Estimated Ultimate Recovery
FDP	Field Development Plan
FID	Final Investment Decision
MOPU	Mobile Offshore Production Unit
OOIP	Original Oil in Place

OFFB-5 Offshore Block-5 (Abu Dhabi)
PIOL Pakistan International Oil Limited
PVT Pressure–Volume–Temperature
UAE United Arab Emirates

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